Lesson 4: Working with Calculations and Methods

Focus: Methods that accept parameters, perform calculations and return values.

The school now wants the system to record students' examination marks and calculate their performance.

Exercise 4 – Adding Student Marks

Modify the `Student` class to include three additional attributes:

- `mathsMark`
- `computerScienceMark`
- `physicsMark`

Update the constructor to initialise these attributes.

Questions

1. Modify the constructor to accept the three subject marks.
2. Add a method called `calculateTotal()` that calculates and returns the total of the three marks.
3. Add a method called `calculateAverage()` that calculates and returns the average mark.
4. Add a method called `getGrade()` that returns a grade based on the student's average.

Use the following grading system:

Average mark	Grade
80–100	A
70–79	B
60–69	C
50–59	D
0–49	E

5. Add a method called `displayResults()` that displays:
    - Student's full name
    - Mathematics mark
    - Computer Science mark
    - Physics mark
    - Total mark
    - Average mark
    - Grade
6. Create three student objects with different marks.

7. Display the examination results for each student.

Challenge questions

8. Add a method called `hasPassed()` that returns `True` if the student's average is 50 or above, otherwise it returns `False`.
9. Add another method called `getPerformanceComment()` that returns:
    - "Excellent performance" for an average of 80 or above.
    - "Good performance" for an average between 60 and 79.
    - "Satisfactory performance" for an average between 50 and 59.
    - "Needs improvement" for an average below 50.

Learning checkpoint: Learners should be able to use methods to perform calculations, return values and make decisions.